

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 10
16 March – 22 March 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



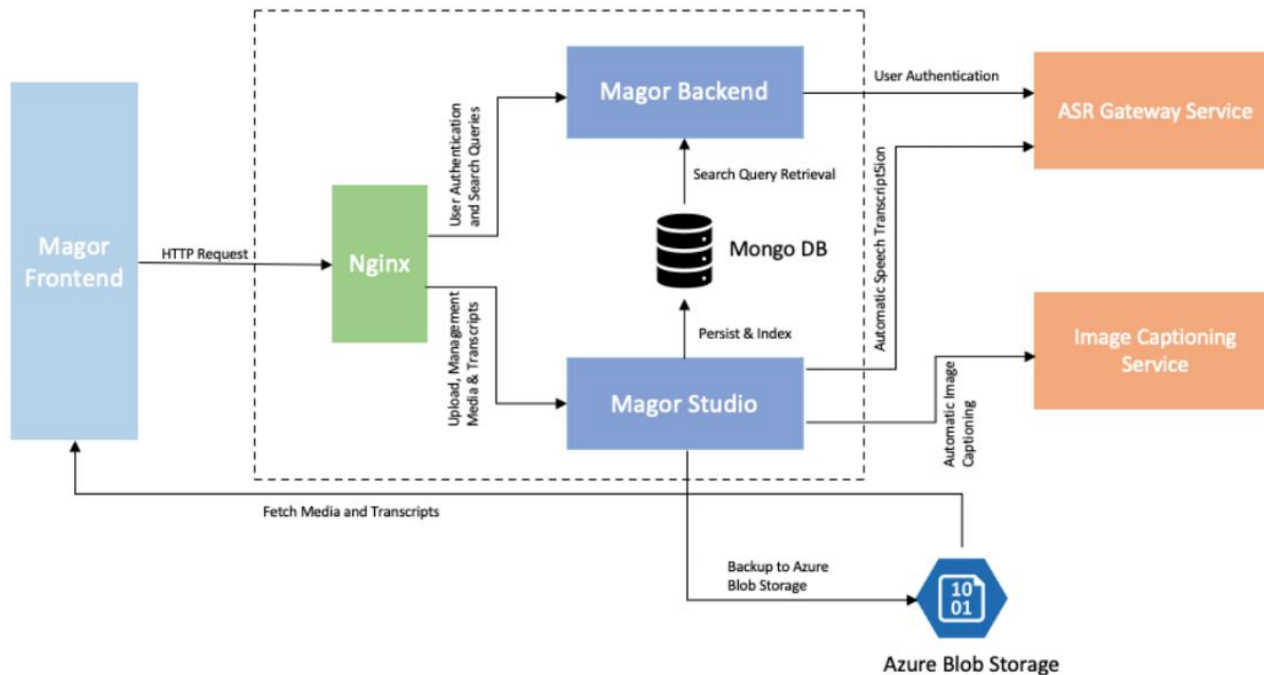
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Admin Recording Edit Page

- Restored retrieval of transcripts
- Fixed field edits error

2. Transcript Studio

- Implemented transcript name display and editing

3. Data Volumes

- Implemented deletion of associated files when deleting recording



Task 1 (Restore Admin Recording Page)

- Restored retrieval of transcripts

The screenshot shows the 'Edit User' interface with a sidebar on the left containing 'Users', 'recordings', 'asrRequests', and 'Statistics'. The main content area is a form for editing a user. The form has a blue header bar with 'Edit User' and a 'BACK TO MAGOR' button. The form fields include: Title (Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4), Description (Podcast by CNA), Recording Path, Transcripts (highlighted with a red box and an arrow pointing to it, with a red text annotation 'Transcripts exist but are not displayed!'), Image Captions (ADD), Sound Captions (Uploader: test), and Upload Date (mm/dd/yyyy). At the bottom, there are 'SAVE' and 'DELETE' buttons.

Before

Task 1 (Restore Admin Recording Page)

- Code Changes:

- Ensures each transcript has a **valid resource path** by checking all possible storage fields.
- Handles inconsistent or legacy transcript formats
 - Filters out nulls or undefined entries
 - Supports string IDs and objects formats
- Maps relevant fields into recording object for frontend-use

```
+ const mapTranscriptEntry = (entry = {}) => ({
+   ...entry,
+   path: entry.path || entry.tempResourceUrl ||
+     entry.azureResourceUrl || '',
+   createdAt: entry.createdAt,
+ })
+
+ const mapTranscriptCollection = (modern = []) => {
+   if (!Array.isArray(modern) || modern.length === 0) return []
+
+   return modern
+     .filter((entry) => entry !== null && entry !== undefined)
+     .map((entry) => {
+       if (typeof entry === 'string') {
+         return mapTranscriptEntry({ _id: entry })
+       }
+       return mapTranscriptEntry(entry)
+     })
+ }
+
+ const normalizeRecording = (record = {}) => ({
+   ...record,
+   transcripts: mapTranscriptCollection(record._transcripts),
+   imageCaptions: mapTranscriptCollection(record._imageCaptions),
+   soundCaptions: mapTranscriptCollection(record._soundCaptions),
+ })
```

Task 1 (Restore Admin Recording Page)

- Restored retrieval of transcripts

The screenshot shows the 'Edit User' interface with a sidebar on the left containing 'Users', 'recordings', 'asrRequests', and 'Statistics'. The main content area has a 'Recording Path' field with the value '/usr/src/app/public/recordings/f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4'. Below this is a 'Transcripts' section, which is highlighted by a red box. This section contains two transcript entries:

Transcript ID	Transcript Path	Transcript name	Uploader	Upload Date	Action
1	/usr/src/app/public/transcripts/d-1773190581977-314209906.xml	test draft	test	03/11/	REMOVE
2	/usr/src/app/public/transcripts/d-1773190581977-314209906-45062655.xml	original	test	03/11/	REMOVE

At the bottom of the 'Transcripts' section is an 'ADD' button. Below the 'Transcripts' section is an 'Image Captions' section, which is partially visible and contains a 'Transcript Path' field and a 'REMOVE' button.

After

Task 1 (Restore Admin Recording Page)

- Recording editing error demo:

Edit User

BACK TO MAGOR

Users

recordings

asrRequests

Statistics

Title

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4

Description

Recording Path

/usr/src/app/public/recordings/f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4

Transcripts

1

↑

↓

Transcript Path

/usr/src/app/public/transcripts/d-1773190581977-314209906.xml

REMOVE

Transcript name

draft test

Uploader

test

Upload Date

03/11/

2

↑

↓

Transcript Path

/usr/src/app/public/transcripts/d-1773190581977-314209906-45062655.xml

REMOVE

Transcript name

original



Task 1 (Restore Admin Recording Page)

Recording editing error

- Cause:
 - Single update handled recording, transcripts, and captions **together, causing inconsistent updates**
- Fix
 - Separation of concern:
 - Recording, transcript and caption updates are now handled through **individual API calls** for reliability

```
259 +   update: async (resource, params) => {
260 +     const isRecording = resource === 'recordings'
261 +     const updateData = isRecording
262 +       ? filterRecordingUpdateData(params.data)
263 +       : params.data
264 +     const transcriptNameUpdates = isRecording
265 +       ? getRecordingTranscriptNameUpdates(params.data,
266 +         params.previousData)
267 +       : []
268 +     const { json } = await
269 +       httpClient(`${apiUrl}/${resource}/${params.id}`, {
270 +         method: 'PATCH',
271 +         body: JSON.stringify(updateData),
272 +       })
273 +     if (isRecording && transcriptNameUpdates.length > 0) {
274 +       await updateTranscriptNames(transcriptNameUpdates)
275 +     }
276 +     return {
277 +       data: {
278 +         ...(isRecording ?
279 +           withRecordingDisplayTranscriptPaths(json) : json),
280 +         id: json._id,
281 +       },
```

Task 1 (Restore Admin Recording Page)

- Recording editing fixed demo:

Edit User

BACK TO MAGOR

Users

recordings

asrRequests

Statistics

Title

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4

Description

testing

Recording Path

/usr/src/app/public/recordings/f3fb4bc3-1f1b-4d7e-abc8-d714e1c448a5.mp4

Transcripts

1

↑ ↓

Transcript Path

/usr/src/app/public/transcripts/d-1773190581977-314209906.xml

REMOVE

Transcript name

test draft

Uploader

test

Upload Date

03/11/

2

↑ ↓

Transcript Path

/usr/src/app/public/transcripts/d-1773190581977-314209906-45062655.xml

REMOVE

Transcript name

original



Task 2 (Transcript Name Display in Transcript Studio)

- Transcript Studio did not display transcript names

The screenshot shows the Transcript Studio interface. At the top, there's a blue header with the name 'Magor' and a dropdown menu showing 'test'. Below the header, the page is titled 'Transcripts' and has a 'Sort By Created Date (Latest)' dropdown. The main content area displays a list of transcripts. The first transcript is highlighted with a red box around its 'Name' input field (containing 'original') and 'Save' button. A red arrow points from the text 'Already implemented in Studio!' to this red box. Below the name field, the transcript ID is '69c20c5936b2c00023c7d95e', and it was created 'a month ago'. The transcript is marked as 'Uploaded'. Below this, there are buttons for 'Default', 'Download', and 'Delete'. The second transcript has the name 'test draft123', ID '69c179fdded95e0023ece3f6', was created '2 months ago', and was last updated 'a day ago'. It also has 'Set Default', 'Download', and 'Delete' buttons. A third transcript is partially visible at the bottom, showing 'Uploaded' status and 'Edit', 'Download', and 'Delete' buttons.



Task 2 (Transcript Name Display in Transcript Studio)

- Transcript Name Edit Demo

Magor

test

Title

Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4

Description

No Description

Go to Recording



ASIA Business Tonight

▶ 🔊

Sort By Created Date (Earliest) ▾

Name: Save

Transcript ID: 69c179fdded95e0023ece3f6 Created At: 12 hours ago Last Updated At: a few seconds ago

Uploaded

Edit

Download ▾

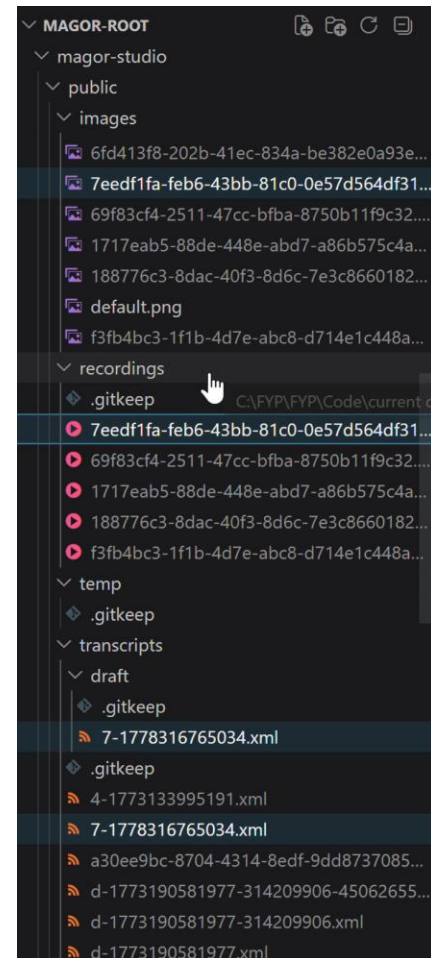
Delete

Name: Save

13

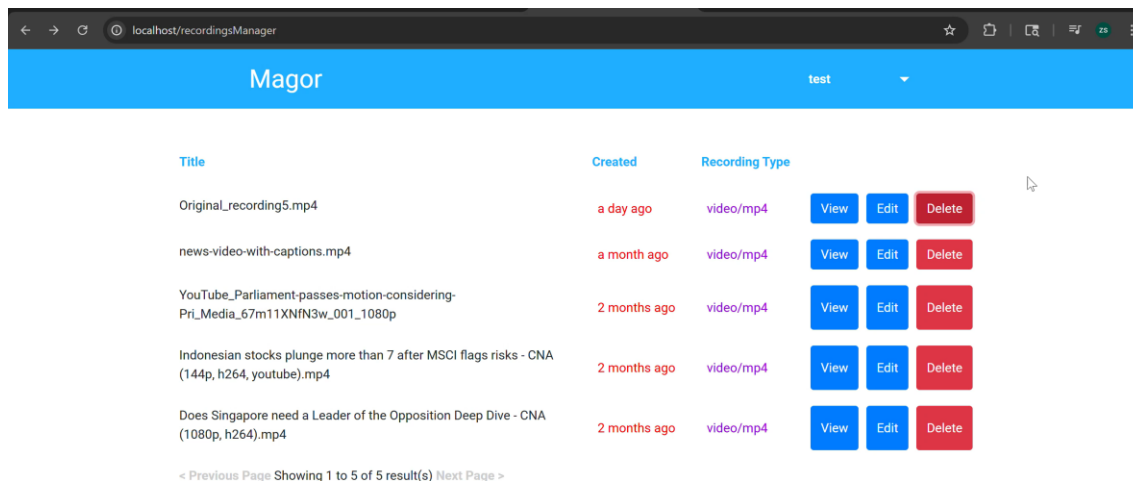
Task 3 (Delete Associated Files When Deleting Recording)

- Recording files were previously stored in Docker volumes (harder to manage and clean up automatically)
- Migrated storage to the local codebase
- **Solution:**
 - Implement automatic file deletion for better space efficiency



Task 3 (Delete Associated Files When Deleting Recording)

- Recording Deletion Demo



The screenshot shows a web browser at localhost/recordingsManager. The application has a blue header with the name 'Magor' and a user profile 'test'. Below the header is a table of recordings. The table has columns for Title, Created, Recording Type, and actions (View, Edit, Delete). There are five rows of recordings, each with a unique title, creation time, and type. The 'Delete' button for each row is highlighted in red. At the bottom of the table, there is a pagination link: '< Previous Page Showing 1 to 5 of 5 result(s) Next Page >'. A mouse cursor is pointing at the 'Delete' button of the first row.

Title	Created	Recording Type	
Original_recording5.mp4	a day ago	video/mp4	View Edit Delete
news-video-with-captions.mp4	a month ago	video/mp4	View Edit Delete
YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	2 months ago	video/mp4	View Edit Delete
Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	2 months ago	video/mp4	View Edit Delete
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	2 months ago	video/mp4	View Edit Delete

< Previous Page Showing 1 to 5 of 5 result(s) Next Page >



Plans for Next Week

- Data Volumes
 - Delete image file also upon recording deletion
- Restore Admin Panel
 - Restore ASR Request Page
 - Restore Statistics Page

